

CHARLEVILLE, Australia

WMO: 945100

Lat: 26.4139S

Lon: 146.2558E

Elev: 995

StdP: 14.18

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.9	35.6	10.8	9.9	73.0	15.4	12.4	68.1	18.8	67.9	17.0	66.2	1.3	170	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	103.0	66.8	100.5	66.4	98.1	66.3	74.7	84.6	73.6	84.1	72.4	83.7	10.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	126.0	77.3	71.1	119.0	76.8	69.5	112.5	76.6	39.1	84.2	37.9	83.6	36.8	83.3	97.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.8	15.2	28.4	107.3	2.4	3.1	26.7	109.5	25.3	111.3	24.0	113.0	22.2	115.2		
(5)			26.3	80.4	3.0	6.4	24.2	85.1	22.5	88.8	20.8	92.4	18.7	97.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.2	84.0	81.6	78.3	70.3	61.4	55.5	54.3	57.8	66.1	72.9	78.0	82.3		
	DBStd	12.14	5.41	5.49	5.49	5.81	6.11	6.14	5.71	6.43	7.15	6.71	6.35	6.00		
	HDD50	44	0	0	0	0	2	16	19	7	0	0	0	0		
	HDD65	1116	0	0	1	20	144	288	335	241	73	13	1	0		
	CDD50	7399	1054	885	878	609	355	181	152	248	482	711	841	1003		
	CDD65	2995	589	465	413	179	32	4	3	17	104	259	392	538		
	CDH74	37862	7901	5821	4787	1936	367	41	48	302	1493	3240	4820	7106		
	CDH80	19613	4586	3150	2374	664	47	1	69	570	1531	2549	4071			
(14) Wind	WSAvg	7.3	8.5	8.4	7.7	6.7	5.9	5.8	6.5	7.1	8.1	8.6	8.6			
(15) Precipitation	PrecAvg	18.7	2.9	2.5	1.9	1.0	1.3	0.9	0.9	0.8	1.0	1.1	1.9	2.2		
	PrecMax	36.1	10.5	11.6	11.2	9.8	7.8	5.4	3.9	3.6	5.0	4.9	11.9	7.7		
	PrecMin	9.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	6.3	2.4	2.5	2.1	1.6	1.6	1.1	1.0	0.8	1.2	1.0	2.2	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.0	105.6	100.7	92.9	84.5	77.6	78.8	86.8	95.2	99.0	104.0	105.9		
		MCWB	68.4	68.9	67.1	65.7	61.8	58.9	56.9	59.0	60.5	62.8	65.2	66.1		
	2%	DB	102.4	100.8	97.0	89.1	80.8	74.9	74.9	80.8	89.8	95.0	99.6	102.4		
		MCWB	67.4	67.5	66.8	64.0	60.2	57.9	55.3	57.0	59.3	62.2	64.3	65.7		
	5%	DB	99.9	97.3	94.1	86.6	78.1	72.0	71.8	77.0	86.4	92.0	96.2	99.0		
		MCWB	67.6	67.5	66.0	62.5	59.0	56.8	54.5	55.2	58.3	61.7	64.3	65.4		
	10%	DB	97.0	94.3	91.3	84.1	75.4	68.7	68.6	73.5	82.8	88.6	92.7	96.3		
		MCWB	67.5	67.3	65.9	61.4	57.6	55.0	53.3	53.9	57.9	60.9	63.4	65.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.8	76.5	74.5	71.5	67.3	63.2	64.8	63.4	67.2	70.1	73.7	74.7		
		MCDB	86.7	85.4	81.8	82.3	73.9	70.0	72.0	76.8	75.9	82.8	80.6	86.5		
	2%	WB	74.7	74.6	72.7	68.7	64.5	61.1	60.5	60.3	64.3	67.7	70.2	73.0		
		MCDB	85.1	84.3	81.8	80.2	73.2	68.6	66.3	73.0	75.9	81.3	81.1	84.2		
	5%	WB	73.5	73.4	71.3	66.8	62.4	59.5	57.8	58.1	62.4	66.1	68.9	71.7		
		MCDB	84.9	83.9	81.8	79.2	72.5	67.4	66.5	71.0	77.0	80.4	81.7	82.9		
	10%	WB	72.2	72.2	69.9	64.8	60.4	57.6	55.2	56.0	60.4	64.3	67.5	70.3		
		MCDB	84.8	83.6	82.0	77.2	71.5	65.7	66.1	69.9	76.8	79.8	82.0	83.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	21.7	22.9	25.0	26.1	24.3	27.0	29.1	28.6	26.8	24.2	23.0		
		MCDBR	25.2	25.9	25.9	27.1	27.3	26.4	29.8	32.0	32.0	30.0	28.3	26.7		
		MCWBR	8.0	8.2	8.8	10.1	11.6	12.4	14.4	14.4	12.9	11.2	9.8	9.3		
		MCDBR	19.1	19.0	18.7	20.3	20.4	19.1	21.5	24.4	24.1	23.1	20.4	20.1		
	5% WB	MCWBR	6.6	6.3	7.2	8.3	9.5	9.9	11.5	12.1	11.4	10.6	8.3	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.360	0.350	0.331	0.313	0.296	0.288	0.283	0.297	0.334	0.347	0.369	0.364		
	taud		2.543	2.582	2.627	2.633	2.621	2.646	2.636	2.567	2.458	2.458	2.448	2.503		
	Ebn at Noon		312	310	307	298	290	286	292	299	300	307	307	311		
	Edh at Noon		35	33	30	27	25	23	24	28	35	37	38	36		
(44) All-Sky Solar Radiation	RadAvg		2348	2199	1956	1651	1323	1120	1261	1574	1915	2207	2305	2359		
	RadStd		125	174	110	106	63	80	89	85	135	166	199	171		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.19	N/A	N/A	+1.58	N/A	N/A	N/A	N/A	+299
(47) Regional (1 neighbor)		+1.19	N/A	N/A	+1.58	N/A	N/A	N/A	N/A	+299

Nomenclature: See separate page